# SuperMart Sales Analytics

## Dataset & Power BI Dashboard Documentation

Project documentation for the SuperMart sales dataset (SuperMart_sales_datatset.xlsx) and the accompanying Power BI dashboard (Dashboard_SuperMart.pbix).

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# 1. Project Overview

This project analyses two years of retail sales for a fictional US supermarket chain, SuperMart. The Excel dataset holds order-line level transactions; the Power BI dashboard turns them into a single interactive page covering sales, profit, quantity, delivery speed, payment behaviour and geography.

Goals:

- Track overall sales, profit and quantity performance across 2019 and 2020.
- Identify the strongest and weakest categories, sub-categories, regions and states.
- Understand customer segments, payment modes and shipping performance.
- Provide an interactive view that can be filtered by region.

# 2. Dataset Description

## 2.1 Summary

| Property | Value |
|---|---|
| File | SuperMart_sales_datatset.xlsx (sheet: Sheet1) |
| Rows x columns | 5,901 x 21 |
| Order period | 1 Jan 2019 to 31 Dec 2020 |
| Unique orders | 3,003 |
| Unique customers | 773 |
| Unique products | 1,755 |
| Geography | United States only; 4 regions, 49 states, 452 cities |
| Grain | One row per product line within an order |

## 2.2 Data Dictionary

| Column | Type | Description |
|---|---|---|
| Row ID | Integer | Row identifier carried over from the source data |
| Order ID | Text | Order reference, e.g. CA-2019-160304 |
| Order Date / Ship Date | Date | Date the order was placed / shipped |
| Ship Mode | Text | Standard Class, Second Class, First Class, Same Day |
| Customer ID / Name | Text | Customer identifier and name |
| Segment | Text | Consumer, Corporate, Home Office |
| Country / State / City | Text | Delivery location |
| Region | Text | Central, East, South, West |
| Product ID / Name | Text | Product identifier and description |
| Category | Text | Furniture, Office Supplies, Technology |
| Sub-Category | Text | 17 sub-categories (e.g. Phones, Chairs, Binders) |
| Sales | Decimal | Line sales value in USD |
| Quantity | Integer | Units sold on the line |
| Profit | Decimal | Line profit in USD (negative = loss) |
| Returns | Decimal | 1 where the line was returned; blank otherwise |
| Payment Mode | Text | COD, Cards, Online |

### 2.3 Data Quality Notes

- No missing values except Returns, which is blank for 5,614 rows; blank means not returned (287 lines flagged as returned).
- No fully duplicated rows. One Row ID value appears twice, so Row ID should not be used as a unique key.
- No ship date precedes its order date. Delivery time ranges from 0 to 8 days.
- 1,098 lines (18.6%) have negative profit.
- Country has a single value (United States), so it carries no analytical variation.

## 3. Key Metrics

| Metric | Value | Note |
|---|---|---|
| Total Sales | $1,565,804 | Sum of Sales |
| Total Profit | $175,262 | Sum of Profit |
| Profit margin | 11.2% | Profit / Sales |
| Total Quantity | 22,317 | Sum of Quantity |
| Average Sale | $265.35 | Per order line |
| Average Quantity per line | 3.78 | |
| Average delivery time | 3.9 days | Ship Date minus Order Date |
| Returned lines | 287 | 4.9% of lines |

## 4. Power BI Dashboard

The report file contains one page ("Page 1", 1280 x 720, fit to page) titled SUPER MART SALES DASHBOARD, with 16 elements: a title text box, a region slicer and 14 visuals. Layout, top to bottom:

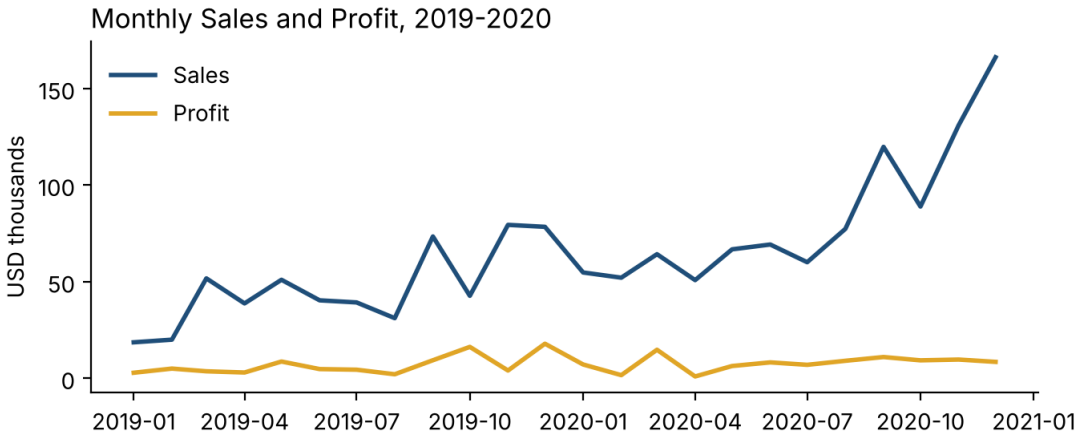
| Visual | Type | Fields used | Purpose |
|---|---|---|---|
| Avg Sale | KPI | Sales | Average sales value |
| Avg Profit | KPI | Profit, Sales | Average profit |
| Avg Quantity / State | KPI | Quantity, State | Average units per state |
| Avg Delivery Days | KPI | AvgDel Time in Days, Sales | Average days from order to ship |
| Quantity / Category | KPI | Category, Quantity | Units per category |
| Quantity / Sub-Category | KPI | Quantity, Sub-Category | Units per sub-category |
| Region slicer | Slicer | Region | Filters the whole page by region |
| Regional Sale | Donut | Region, Sales | Sales share by region |
| Sale via Payment Mode | Donut | Payment Mode, Sales | Sales share by COD / Cards / Online |
| Sales by Segment | Donut | Segment, Sales | Sales share by customer segment |
| Sales trend | Area | Order Date, Sales | Sales over time |
| Monthly Profit by YoY | Area | Order Date, Profit | Profit over time, year on year |
| Sales by Ship Mode | Clustered bar | Ship Mode, Sales | Sales per shipping method |
| Sales by Category | Column | Category, Sales | Sales per category |
| Sales by Sub Category | Funnel | Sub-Category, Sales | Ranking of sub-categories |
| Sales and Profit by State | Map | State, Location, Sales, Profit | Geographic distribution |

**Model note:** the dashboard uses a calculated field named "AvgDel Time in Days" (average of Ship Date minus Order Date) that does not exist in the raw Excel file. The data model inside the .pbix is stored in a compressed binary format, so its exact DAX expressions are not reproduced here; open the file in Power BI Desktop (Model view / Transform Data) to inspect them.

# 5. Insights

## 5.1 Overall performance

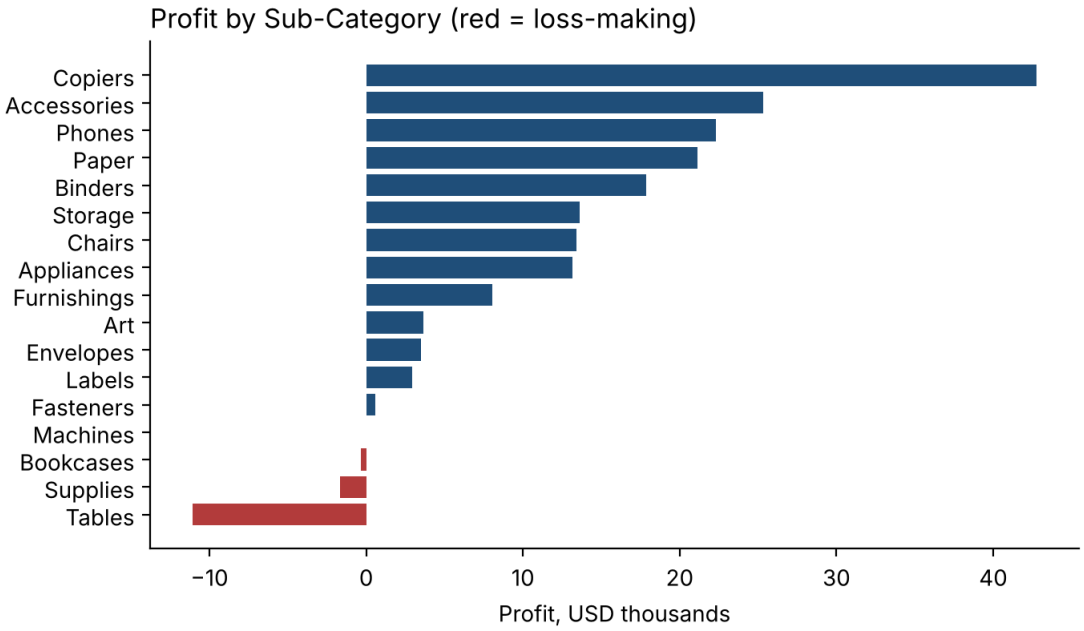
- Sales grew 77% from $564,680 in 2019 to $1,001,125 in 2020, but profit rose only 14% ($81,823 to $93,439). Margin fell from 14.5% to 9.3%.
- December 2020 was the peak month at $166,186 in sales.




## 5.2 Category and sub-category

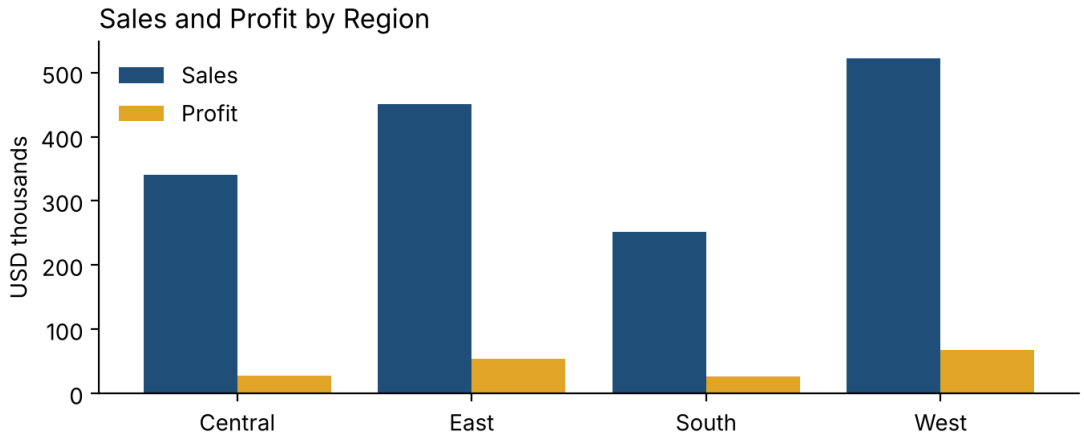
| Category | Sales | Profit | Margin | Quantity |
|---|---|---|---|---|
| Furniture | $451,509 | $10,007 | 2.2% | 4,631 |
| Office Supplies | $643,708 | $74,797 | 11.6% | 13,625 |
| Technology | $470,588 | $90,458 | 19.2% | 4,061 |

- Technology is the most profitable category despite lower volume; Furniture earns almost no profit.
- Tables (-$11,092), Supplies (-$1,654) and Bookcases (-$343) lose money. Copiers have the best profit ($42,775 on $59,736 sales).
- Top sales sub-categories: Phones ($196,564), Chairs ($181,946), Binders ($174,978).




### 5.3 Region and geography

- West leads in sales ($522,441) and margin (13.0%); Central is weakest on margin (8.0%).
- Top states by sales: California ($335,190), New York ($186,748), Texas ($116,262), Washington ($92,975), Pennsylvania ($82,355).




### 5.4 Customers, payments and shipping

- Consumer is the largest segment ($753,002 sales), then Corporate ($509,743) and Home Office ($303,059).
- Cash on delivery accounts for 43% of sales ($667,418), ahead of Online ($553,993) and Cards ($344,393).
- Standard Class carries 58% of sales ($912,401); Same Day is the smallest ($95,959).
- Top customer: Caroline Jumper ($11,597). Top product: 3D Systems Cube Printer, 2nd Generation, Magenta ($14,335).

## 6. Recommendations

- Review pricing and discounting on Tables, Bookcases and Supplies, which sell but lose money.
- Investigate why margin dropped in 2020 even as sales grew strongly.
- Lift Central region profitability, and grow Technology, the highest-margin category.
- Review COD handling costs and returns, given COD is the largest payment channel.

## 7. How to Use

- Download both files and open Dashboard_SuperMart.pbix in Power BI Desktop (Windows).
- If prompted, update the data source path to your local copy of the Excel file (Home > Transform data > Data source settings).
- Use the Region slicer at the top right to filter every visual.

## 8. Tools & Technologies

Microsoft Excel (data source), Microsoft Power BI Desktop (modelling and dashboard), Python / pandas (documentation analysis).